

# Predictive Analysis Project

## Football Player Performance Analysis

Your Name

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# Objective

- Predict football player goals
- Identify high scorers
- Compare statistical and ML models

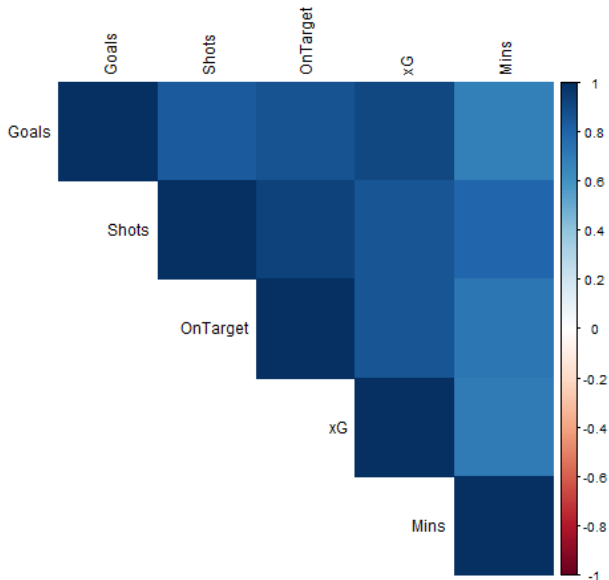
This project analyzes football performance using regression and classification techniques.

# Data Description

- Dataset: Football players
- Variables: Goals, Shots, OnTarget, xG, Minutes

Each row represents a player's match performance statistics.

# Correlation Analysis



# Linear Regression

```
model1 = lm(Goals ~ Shots, data=df)
```

# Linear Regression Summary

- $R\text{-squared} = 0.697$
- Shot coefficient = 0.143
- Highly significant ( $p < 2e-16$ )

Shots positively influence goals but model is simple.

# Multiple Regression

```
model2 = lm(Goals ~ Shots + OnTarget + xG, data=df)
```

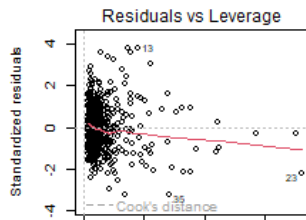
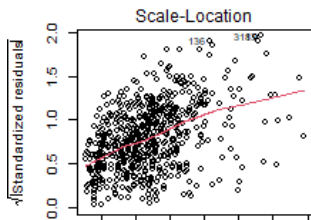
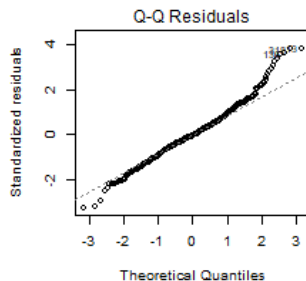
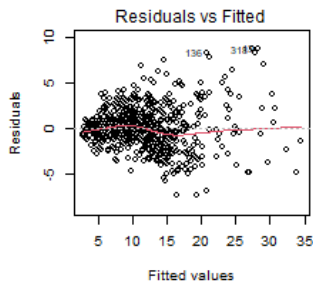
# Multiple Regression Summary

- $R\text{-squared} = 0.852$
- $xG = 0.661$  (strongest factor)
- $OnTarget = 0.133$  (significant)
- $Shots = -0.007$  (not significant)

$xG$  is the most important predictor of goals.



# Regression Diagnostics



# Diagnostic Interpretation

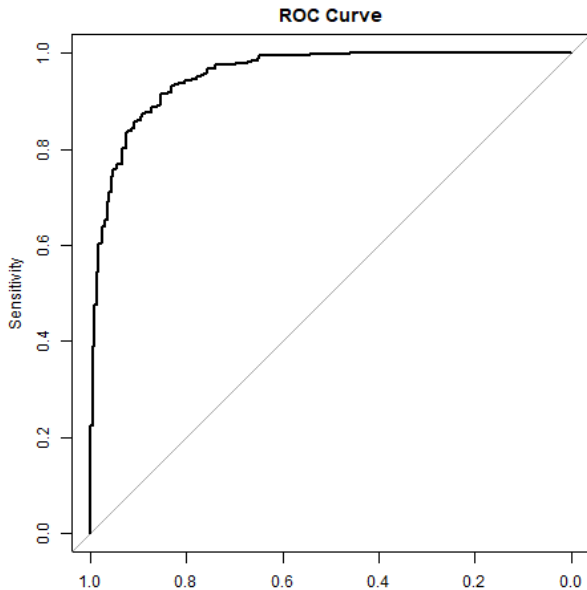
- Residuals randomly scattered
- No pattern  $\rightarrow$  good model fit
- No major outliers
- Assumptions satisfied

Model is statistically reliable.

# Logistic Regression

```
glm(HighScorer ~ Shots + xG, family="binomial")
```

# ROC Curve



# ROC and AUC

$$TPR = \frac{TP}{TP + FN} \quad FPR = \frac{FP}{FP + TN}$$

AUC measures overall classification ability of the model.

# AUC Interpretation

- 0.5 = random model
- 1 = perfect model
- Higher value = better performance

# Confusion Matrix

|             | Actual 0 | Actual 1 |
|-------------|----------|----------|
| Predicted 0 | 190      | 29       |
| Predicted 1 | 40       | 401      |

Most predictions are correct with few errors.

# Model Performance

- Accuracy = 89.55%
- Sensitivity = 0.826
- Specificity = 0.933
- Precision = 0.868
- Kappa = 0.767



# F1 Score

$$F1 = 2 \times \frac{\textit{Precision} \times \textit{Recall}}{\textit{Precision} + \textit{Recall}}$$

F1 balances precision and recall.

# Lasso Regression

```
cv.lasso = cv.glmnet(x, y, alpha=1)
```

# Lasso Interpretation

Lasso removes weak variables and keeps only important predictors, reducing overfitting and improving simplicity.

# Model Comparison

| Model    | Purpose           | Strength        |
|----------|-------------------|-----------------|
| Linear   | Prediction        | Simple          |
| Multiple | Prediction        | Strong accuracy |
| Logistic | Classification    | Good separation |
| Lasso    | Feature selection | Reduces noise   |

# Conclusion

- xG is the strongest predictor
- Multiple regression explains 85
- Logistic regression gives 90
- Lasso improves model efficiency

Statistical models effectively analyze football performance data.